Name of Course	**Bachelor of Computer Application (BCA)**
Semester	**VI**
Name of Subject	**Cyber Security**
Subject Code	**BCA-605-B**
Marks	**75 Marks**
Lectures	**50 Lectures**

Unit I

1. IT Act and Encryption

1.1 Object of the Act
1.2 Scope of the Act
1.3 Symmetric Cryptography
1.4 Asymmetric Cryptography
1.5 RSA Algorithm
1.6 Public Key Encryption

1.1 Object of the Act:

The **object of the Act** in cyber security—mainly referring to laws like the **Information Technology (IT) Act, 2000** and its amendments—is to provide a **legal framework** for regulating activities in the digital environment. Its primary objective is to ensure **safe, secure, and reliable use of computers, networks, and electronic transactions**.

The key objects include:

1. **Legal Recognition of Electronic Records and Digital Signatures:**
   The Act aims to promote electronic communication and e-commerce by giving legal validity to digital documents and electronic signatures.

2. **Protection Against Cybercrimes:**
   It defines various cyber offences such as hacking, identity theft, cyberstalking, and unauthorized access, and prescribes penalties to deter such activities.

3. **Safeguarding Privacy and Data Security:**
   The Act sets guidelines for protecting sensitive personal data and ensuring confidentiality, integrity, and availability of information systems.

4. **Regulation of Intermediaries and Cyber Entities:**
   It outlines responsibilities of service providers, social media platforms, and intermediaries to prevent misuse of their platforms and to cooperate with law enforcement agencies.

5. **Promoting Trust in Digital Transactions:**
   By enforcing cyber laws and securing online systems, the Act builds user confidence in e-governance, e-banking, and digital services.

In short:

The object of the Act in cyber security is **to create a secure digital ecosystem, prevent**

cybercrimes, protect data, regulate online activities, and promote safe electronic transactions.

1.2 Scope of the Act:

The **scope of the Information Technology (IT) Act, 2000** defines the range of activities, sectors, and digital operations that the law covers. It applies to all electronic records, digital transactions, and cyber activities occurring within India or affecting Indian systems. The main points of its scope are:

1. **Applicability to Electronic Records and Digital Signatures:**
   The Act governs the creation, storage, transmission, and authentication of electronic records and digital signatures, giving them legal recognition.

2. **Coverage of Cybercrimes and Offences:**
   It includes offences such as hacking, data theft, cyberstalking, identity theft, phishing, and unauthorized access. It applies to individuals, companies, and organizations.

3. **Regulation of Intermediaries and Cyber Service Providers:**
   The Act covers ISPs, social media platforms, e-commerce companies, and network service providers, defining their obligations and liability.

4. **Jurisdiction Beyond Physical Boundaries:**
   The Act applies to cyber offences committed **within India or outside India** if the offence involves a computer or network located in India.

5. **E-Governance and Digital Transactions:**
   It covers electronic contracts, e-filing, e-payments, and digital communication with government agencies, promoting secure and efficient e-governance.

In short:

The scope of the Act extends to **all electronic records, digital transactions, cyber offences, intermediaries, and computer-related activities**, even if performed outside India but impacting Indian systems.

1.3 Symmetric Cryptography:

Symmetric cryptography, also called **secret-key cryptography**, is an encryption technique where the **same key** is used for both **encryption and decryption** of data. It is one of the oldest and fastest methods of securing information.

Key Features:

1. **Single Key Usage:**
   A common secret key is shared between the sender and receiver. The security of the system depends on keeping this key confidential.

2. **High Speed and Efficiency:**
   Symmetric algorithms are computationally faster, making them suitable for encrypting large amounts of data such as files, databases, and communication streams.

3. **Common Algorithms:**
   Popular symmetric algorithms include **AES (Advanced Encryption Standard)**, **DES**

(Data Encryption Standard), and **Blowfish**.

4. **Applications:**
   Used in secure communication, data storage, VPNs, and wireless security. It is widely used in real-time systems requiring quick data processing.

5. **Limitation – Key Distribution Problem:**
   The major challenge is securely sharing the secret key between sender and receiver. If the key is exposed, the entire communication becomes insecure.

In short:
Symmetric cryptography is a fast and efficient method of encryption that uses a **single shared secret key**, but faces challenges in **key distribution**.

1.4 Asymmetric Cryptography:

Asymmetric cryptography, also known as **public-key cryptography**, is an encryption technique that uses **two different keys**: a **public key** for encryption and a **private key** for decryption. These keys are mathematically related but cannot be derived from each other easily, which ensures strong security.

Key Features:

1. **Two-Key System:**
   Each user has a **public key** (shared with everyone) and a **private key** (kept secret). Data encrypted with one key can only be decrypted using the other.

2. **High Security for Key Exchange:**
   It solves the **key distribution problem** of symmetric cryptography. The sender does not need to share a secret key; they can use the receiver's public key.

3. **Digital Signatures:**
   Asymmetric cryptography enables authentication and non-repudiation using digital signatures. A message signed with a private key can be verified using the corresponding public key.

4. **Common Algorithms:**
   Popular algorithms include **RSA (Rivest–Shamir–Adleman)**, **DSA (Digital Signature Algorithm)**, and **Elliptic Curve Cryptography (ECC)**.

5. **Applications:**
   Used in secure email (PGP), SSL/TLS for website security, digital certificates, blockchain, and secure key exchange in many protocols.

In short:
Asymmetric cryptography uses a **public key–private key pair** to provide secure communication, authentication, and digital signatures, solving the limitations of symmetric key exchange.

1.5 RSA Algorithm:

RSA (Rivest–Shamir–Adleman) is one of the most widely used **asymmetric cryptographic algorithms**. It uses a **public key** for encryption and a **private key** for decryption. The security of RSA is based

on the mathematical difficulty of **factorizing large prime numbers**.

Steps of the RSA Algorithm:

1. **Key Generation:**
   Select two large prime numbers **p** and **q**.
   Compute $\mathbf{n = p \times q}$, which becomes part of the public and private key.
   Calculate $\boldsymbol{\varphi(n) = (p - 1)(q - 1)}$.
   Choose a public key **e** such that it is relatively prime to $\varphi(n)$.
   Compute the private key **d**, where $\mathbf{d \times e \equiv 1 \pmod{\varphi(n)}}$.

2. **Encryption:**
   The sender uses the **receiver's public key (e, n)**.
   Ciphertext is computed as:
   $$\mathbf{C = M^e \bmod n},$$
   where **M** is the plaintext message.

3. **Decryption:**
   The receiver uses their **private key (d, n)**.
   Plaintext is recovered as:
   $$\mathbf{M = C^d \bmod n}.$$

4. **Security:**
   RSA is secure because factoring **n** into its prime components (p and q) is extremely difficult for large key sizes (1024–4096 bits).

5. **Applications:**
   Used in **digital signatures**, **SSL/TLS**, **email encryption (PGP)**, and **secure key exchange**.

In short:

The RSA algorithm is a **public-key cryptographic scheme** that provides secure encryption and digital signatures by using **two keys** and the complexity of factoring large primes.

1.6 Public Key Encryption:

Public Key Encryption, also known as **asymmetric encryption**, is a cryptographic technique where two different keys are used for secure communication: a **public key** for encryption and a **private key** for decryption. The keys are mathematically related but cannot be derived from each other easily, which ensures strong security.

Key Points:

1. **Two-Key System:**
   Each user has a **public key** (shared openly) and a **private key** (kept secret). Anyone can encrypt data using the public key, but only the owner can decrypt it using the private key.

2. **Secure Key Distribution:**
   Unlike symmetric encryption, there is **no need to share a secret key** beforehand. This eliminates the key distribution problem and supports secure communication over open networks.

3. **Confidentiality and Authentication:**
   Encryption with a public key ensures **confidentiality**.
   When data is encrypted or signed using a private key, it provides **authentication** and **non-repudiation** (digital signatures).

4. **Common Algorithms:**
   Public key encryption uses algorithms like **RSA**, **Elliptic Curve Cryptography (ECC)**, and **ElGamal**.

5. **Applications:**
   Used in **SSL/TLS for website security**, **email encryption**, **digital certificates**, **blockchain**, and **secure key exchange** in communication protocols.

In short:

Public key encryption uses a **public–private key pair** to provide **secure communication, authentication, and digital signatures**, making it suitable for use in modern internet and cybersecurity applications.